

Math 55 Final Exam Solutions.

2.1. False. Taking $S = \{x \in \mathbb{Q} : x < 0\}$ we see that S has no largest element.

2.2. True. If $p \rightarrow q$ is false then we must have $p = T$ and $q = F$, in which case $\neg p \rightarrow \neg q$ is true.

2.3. True. Assume $(a, b) \in A \times B$. Then $a \in A$ so $a \in B$ since $A \subseteq B$. Thus, $(a, b) \in B \times B$, so $A \times B \subseteq B \times B$.

2.4. False. Consider $A = \{1\}$, $P(A) = \{\emptyset, \{1\}\}$, and $B = \mathbb{R}$. Then it is easy to find a one to one function $f : P(A) \rightarrow \mathbb{R}$ but $\mathbb{R}$ is uncountable.

2.5. True. Assume that A is infinite, and assume for contradiction that B is countable and $f : A \rightarrow B$ is an injection (one to one). Thus, there is an injection $g : \mathbb{Z} \rightarrow A$, which means there is an injection $h : P(\mathbb{Z}) \rightarrow P(A)$. This implies that $f \circ h : P(\mathbb{Z}) \rightarrow B$ is an injection, which is a contradiction since B is countable and $P(\mathbb{Z})$ is uncountable.

2.6. False as stated: Consider the uncountable set of one-vertex graphs with $V = \{x\}$ where x is a real number. But the intention of the question was to ask about graphs with vertex set $V = \{1, \dots, n\}$ for some positive integer n . The set of such graphs is countable: we simply list the graphs on one vertex, then the ones on two vertices, and in general the ones on n vertices; this is possible since the number of simple graphs on n vertices with vertex set $\{1, \dots, n\}$ is $2^{\binom{n}{2}}$, which is finite.

2.7. True. If $\gcd(a^2, b^3) > 1$ then there is some k such that $k|a^2$ and $k|b^3$. By Euclid's lemma (or the fundamental theorem of arithmetic) this implies that $k|a$ and $k|b$, so $\gcd(a, b) > 1$, and they cannot be relatively prime.

2.8. True. $P(E) = 13/52 = 1/4$, $P(F) = 4/52 = 1/13$, and $P(E \cap F) = 1/52 = P(E)P(F)$.

2.9. False. Consider the triangle graph, which has three vertices of degree equal to 2. *In contrast, the number of vertices of odd degree must be even, by the handshaking lemma.*

2.10. False. Recall that a graph is Eulerian if and only if all of its vertices have even degree. So all we need to do is find a graph which has an odd degree vertex, and whose complement also has an odd degree vertex. If G has n vertices, every vertex x satisfies $\deg_G(x) + \deg_{\overline{G}}(x) = n - 1$ since the complete graph has degree $n - 1$. Choosing $n = 3$ and taking G to be the path graph on 3 vertices, we find that $\overline{G}$ consists of a single edge and an isolated vertex, and neither graph has an Eulerian circuit.

3. Observe that if x satisfies the equation then $x \not\equiv 0 \pmod{5}$ since otherwise we would have $4x^{21} \equiv 0 \pmod{5}$. By Fermat's little theorem, we therefore have $x^{5-1} = x^4 \equiv 1 \pmod{5}$, so the equation simplifies to

$$4x^{21} \equiv 4(x^4)^5 \cdot x \equiv 4x \equiv 2 \pmod{5}$$

. Since $4 \equiv -1 \pmod{5}$ this further simplifies to $x \equiv -2 \pmod{5}$, so $x = -2$ (or any x congruent to $-2 \pmod{5}$, such as $x = 3, 8, \dots$) is a solution to the equation.

4. The second equation is equivalent to $6 \mid 2x$, which implies $2x = 6k$ for some integer k or $x = 3k$, i.e. $3 \mid x$. This contradicts the first equation, so no such integer x exists.

5. (a) Let $[n] = \{1, \dots, n\}$ be the students. Let's calculate $\mathbb{E}T_n$ by using linearity of expectation to break it into simpler random variables. For every (unordered) subset $S \subseteq [n]$ with $|S| = 3$ define an indicator random variable X_S , equal to one when the three students in S have the same birthday; there are $\binom{n}{3}$ such random variables. We then have

$$T_n = \sum_{S \subseteq [n], |S|=3} X_S$$

so

$$\mathbb{E}T_n = \sum_{S \subseteq [n], |S|=3} \mathbb{E}X_S$$

by linearity of expectation. For a fixed S ,

$$\mathbb{E}X_S = \mathbb{P}(\text{the three students in } S \text{ have the same birthday}) = \frac{365}{365^3} = \frac{1}{365^2}.$$

Thus,

$$\mathbb{E}T_n = \sum_{S \subseteq [n], |S|=3} \frac{1}{365^2} = \frac{n(n-1)(n-2)}{3 \cdot 2} \frac{1}{365^2}.$$

Plugging in $n = 120$ we have

$$\mathbb{E}T_n = \frac{120 \cdot 119 \cdot 118}{6 \cdot 365^2} \geq \frac{20 \cdot 118^2}{365^2}.$$

Since $118/365 > 1/4$ this fraction is at least $20/16 > 1$, as desired.

(b) By the pigeonhole principle, if $n \geq 2 \cdot 365 + 1 = 731$, then there must be some birthday with 3 students. This is the minimum such number since for $n = 730$ we can have exactly two students for each of the 365 birthdays, and no triples.

6. Let X_j be the indicator random variable equal to 1 if the j^{th} coin is heads, $j = 1, \dots, n$. Note that

$$\mathbb{E}X_j = \mathbb{P}(j^{\text{th}} \text{ coin is heads}) = j/n.$$

(a) By linearity of expectation

$$\mathbb{E}H_n = \sum_{j=1}^n \mathbb{E}X_j = \sum_{j=1}^n (j/n) = 1/n \sum_{j=1}^n j = n \cdot \frac{n(n+1)}{2} = \frac{n+1}{2}.$$

(b) Since X_j is an indicator random variable,

$$\mathbb{E}X_j^2 = \mathbb{E}X_j = j/n.$$

Thus,

$$V(X_j) = \mathbb{E}X_j^2 - (\mathbb{E}X_j)^2 = (j/n) - (j/n)^2.$$

Since the X_j are independent,

$$V(H_n) = \sum_{j=1}^n V(X_j) = \sum_{j=1}^n (j/n) + \sum_{j=1}^n (j/n)^2 = \frac{n+1}{2} - \frac{1}{n^2} \sum_{j=1}^n j^2 = \frac{n+1}{2} - \frac{1}{n^2} \frac{n(n+1)(2n+1)}{6}$$

by the formula for the sum of the squares of the first n integers.

(c) By Markov's inequality

$$P(H_n \geq 9n/10) \leq \frac{\mathbb{E}H_n}{(9n/10)} = \frac{10(n+1)}{2 \cdot 9n} = \frac{5n+5}{9n}.$$

7. Suppose $G = (V, E)$ has n vertices and at least n edges. Assume for contradiction that G is acyclic. Let $G_1, \dots, G_k$ be the connected components of G for an integer $k \geq 1$, with vertex sets of sizes $n_1, \dots, n_k$ such that $\sum_{i=1}^k n_i = n$. Since each G_i is acyclic (being a subgraph of G) and connected (being a connected component), each G_i is a tree, which must have exactly $n_i - 1$ edges by a theorem proved in class. Thus the total number of edges of G is

$$|E| = \sum_{i=1}^k (n_i - 1) = \left(\sum_{i=1}^k n_i \right) - k = n - k,$$

where $k \geq 1$. In particular $|E| < n$, contradicting that $|E| = n$ and completing the proof.

This problem can also be done by induction.

8.1. (a) We proceed by induction. The claim is true for $n = 1$ vertices. Let G be a tree on n vertices. Let x be a leaf of G , and let $G' = G - \{x\}$ be the tree obtained by deleting x and its incident edge $e = xy$, where $y \in G'$; note that G' has $n - 1$ vertices. Observe that a 2-coloring f

of G is uniquely determined by the following process: (1) choose a 2-coloring of G' (2 ways, by the inductive hypothesis applied to G') (2) choose a color for x (one way: the opposite of the color chosen for x). Since there are 2×1 ways to make these choices, there must be exactly two 2-colorings of G , as desired.

8.2. (a) Let a_n denote the number of 3-colorings of a path graph with n vertices (the case $n = 1$ corresponding to a single vertex). The base case is $n = 1$ for which $a_1 = 3$. If G is a path on n vertices, then a 3-coloring is uniquely specified by the following process: (1) choose a 3-coloring of $G' = G - \{x\}$ where x is the last vertex (leaf). There are a_{n-1} ways to do this. (2) Choose a color for x , distinct from its neighbor in G' . There are 2 ways to do this. Thus we have the recurrence

$$a_n = 2a_{n-1} \quad a_1 = 3,$$

which has the solution $a_n = 3 \cdot 2^{n-1}$.

(b) This is very similar to 8.1a and 8.2a, but with a slightly more complicated induction hypothesis since the number of k -colorings depends on k . The quickest way is to guess what this number of colorings might be, based on 8.2a: if we considered k -colorings of the path rather than 3-colorings then the same argument would give $a_n = k \cdot (k-1)^{n-1}$. If all trees on n vertices have the same number of k -colorings, then they must have the same number as the path. So let the inductive hypothesis be "All trees on n vertices have $k \cdot (k-1)^{n-1}$ k -colorings."

The basis step $n = 1$ is clear since a single vertex has exactly k k -colorings. For the inductive step, let G be a tree on n vertices. Let x be a leaf and let $G' = G - \{x\}$. Every k -coloring of G is uniquely determined by the following process: (1) Choose a k -coloring of G' (there are $k \cdot (k-1)^{n-2}$ ways to do this by the inductive hypothesis since G' has $n-1$ vertices) (2) Choose a color for x distinct from the color of its neighbor in G' (there are $k-1$ ways to do this). Thus, the total number of k -colorings is $k \cdot (k-1)^{n-2} \cdot (k-1) = k \cdot (k-1)^{n-1}$, completing the induction.

Alternatively you could derive a recurrence for the number of k -colorings and solve it, but this is a bit trickier to write since you don't know at the outset that every tree on n vertices has the same number of k -colorings.

9. (a) Let n be even and let $G = (V, E)$ be a graph on n vertices, with $\deg(x) \geq n/2$ for every vertex $x \in V$. We will show that G is connected. Suppose $a, b \in V$ are two arbitrary distinct vertices. If a and b are adjacent then there is a path of length 1 between them. Otherwise, let

$S = V - \{a, b\}$ be the set of vertices of G other than a and b . Let A be the set of neighbors of a and let B be the set of neighbors of b . Notice that $A \subseteq S$ and $B \subseteq S$ since a, b are not adjacent. Observe that

$$|A| + |B| \geq \deg(a) + \deg(b) \geq n/2 + n/2 = n,$$

and moreover $|A \cup B| \leq |S| = n - 2$, so A and B cannot be disjoint (this can also be seen by the formula $|A \cup B| = |A| + |B| - |A \cap B|$). Thus, there is a vertex $c \in A \cap B$, and there must be a path a, c, b between a and b , as desired.

(b) Let G be a disjoint union of two complete graphs on $n/2$ vertices. Every vertex in G has degree exactly $n/2 - 1$ (since the degree of a vertex in a complete graph is one less than the number of vertices), but G is disconnected.